

CS5700

Final Project and Report Total

(100 marks)

As a part of this assignment, students are expected to build a networking application. Your final project should be a development project related to a networking application. You can work in a group of 2-4 members.

Students are expected to conduct a presentation on the same as well as submit a report.

The presentation and report should cover the following sections (10 Marks each) (not limited to):

- Introduction
- System features
- System analysis and design.
- Implementation details and methodology
- Testing and user manual
- Any improvements and future work
- Conclusion
- Reflection

Total: 70 marks for the report with the related implementation of the project.

All sections must be explained with sufficient detail and clarity. Include screenshots and references wherever necessary.

Distribution of marks:

Clear presentation, slides etc.: 20 Marks

Q&A: 10 Marks

Presentation: Max: 7 minutes.

Q&A: 3 minutes

Submission will include a set of slides and detailed report not exceeding 2500

words. Source code should be uploaded into GitHub.
You group can choose any one of the following projects.

Project 1:

For the purpose of this assignment, students are expected to implement the following:

The goal of this project is to design and implement a networking application that facilitates communication and data exchange between users or systems. The application will feature a user interface with real-time networking capabilities. The designed project should have the following features:

- 1) Enable instant messaging: Support for text, voice, or video communication
- 2) File sharing
- 3) Communication should be secure
- 4) Secure login with options for multi-factor authentication
- 5) Support for creating and managing group chats or collaborative sessions
- 6) Logs or dashboards for tracking connection activity
- 7) Intuitive and user-friendly front end

Project 2:

For the purpose of this assignment, students are expected to implement the following:

This project involves developing a multiplayer Tic-Tac-Toe game where two players can connect over a network and play in real time. The application will feature a user interface with real-time networking capabilities.

The designed project should have the following features:

- 1) Players take turns and moves update instantly.
- 2) Users can join and wait for an opponent.
- 3) Ensure both players see the same board.
- 4) Players can message each other while playing
- 5) Track wins, losses, and draws
- 6) Logs or dashboards for tracking connection activity
- 7) Intuitive and user-friendly front end

Project 3:

For the purpose of this assignment, students are expected to implement the following:

This project involves developing a Smart Weather Monitoring System, where an IoT device collects real-time environmental data (temperature, humidity, and air quality) and transmits it to a remote server. Users can access the data via a web application or mobile application.

The designed project should have the following features:

- 1) Measure temperature or humidity, and/ or air quality.
- 2) Users can view data on a web or mobile dashboard.
- 3) Data should be logged in a remote server to store historical data for trend analysis.
- 4) Send alerts for extreme weather conditions to the user
- 5) Intuitive and user-friendly front end

Project 4:

For the purpose of this assignment, students are expected to implement the following:

This project involves developing a Smart Home IoT Light Control System, where users can remotely turn lights ON or OFF using a web or mobile application. The system communicates with a server over a network (Wi-Fi) to control the light.

The designed project should have the following features:

- 1) Users can toggle the light ON/OFF from anywhere.
- 2) The application shows the current state of the light in real time.
- 3) Only authenticated users should be able to access the system.
- 4) Users can set a timer for the light.
- 5) Intuitive and user-friendly front end